

Round 2 Evaluation Metrics Report

Dataset and evaluation design

The dataset contains 9,000 labelled text records. Two supervised NLP tasks were evaluated:

1. Sentiment classification.
2. Topic-category classification.

A group-aware split prevented identical text values from appearing in more than one data split.

Split	Records
Training	5382
Validation	1805
Testing	1813

Final metrics were calculated only on the untouched test set.

1. Sentiment classification

Target

```
post_text → sentiment_label
```

The classes are Negative, Neutral, and Positive.

Model selection

Candidate model	Accuracy	Macro F1	Weighted F1
word_C1.0_cwNone	0.6183	0.6169	0.6170
word_C2.0_cwbalanced	0.6127	0.6119	0.6119
char_C1.0_cwbalanced	0.6039	0.6031	0.6031
combined_C0.5_cwNone	0.6133	0.6120	0.6121
combined_C1.0_cwNone	0.6183	0.6169	0.6170
combined_C2.0_cwNone	0.6133	0.6125	0.6125

Selected model: **combined_tfidf_logistic_regression**

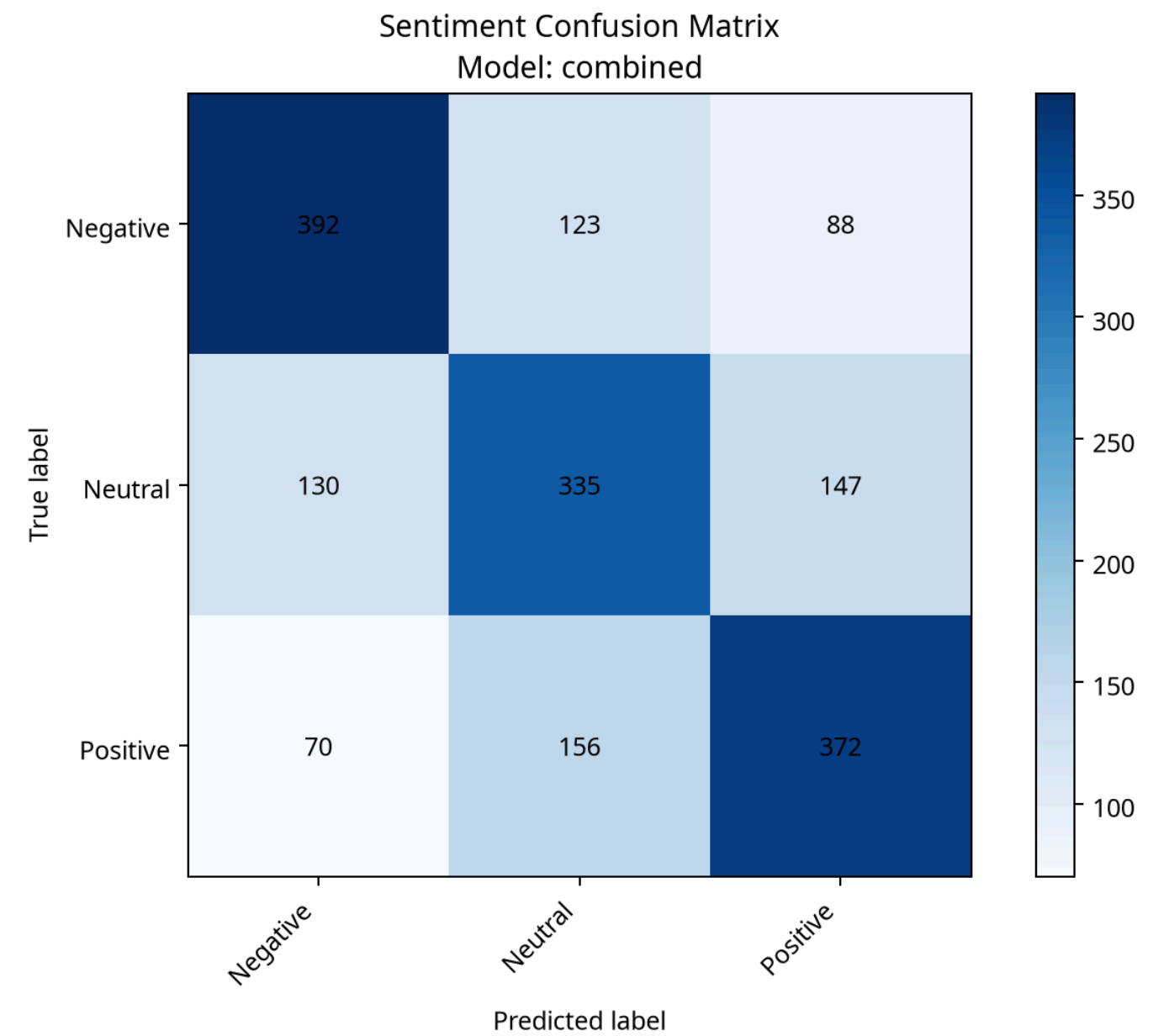
Final test metrics

Metric	Score
Accuracy	0.6062
Macro precision	0.6069
Macro recall	0.6065
Macro F1-score	0.6067
Weighted F1-score	0.6063

Per-class metrics

Class	Precision	Recall	F1-score	Support
Negative	0.6622	0.6501	0.6561	603
Neutral	0.5456	0.5474	0.5465	612
Positive	0.6129	0.6221	0.6174	598

Confusion matrix



2. Topic classification

Target

```
post_text → topic_category
```

The topic classes are Account Security, Community Discussion, Feature Feedback, and Technical Issues.

Model selection

Candidate model	Accuracy	Macro F1	Weighted F1
word_C1.0_cwbalanced	0.9291	0.6294	0.9138
char_C0.5_cwbalanced	0.9490	0.7315	0.9407
char_C1.0_cwbalanced	0.9457	0.6933	0.9344
combined_C1.0_cwbalanced	0.9291	0.6294	0.9138
combined_C2.0_cwbalanced	0.9307	0.6325	0.9155

Selected model: **char_tfidf_logistic_regression**

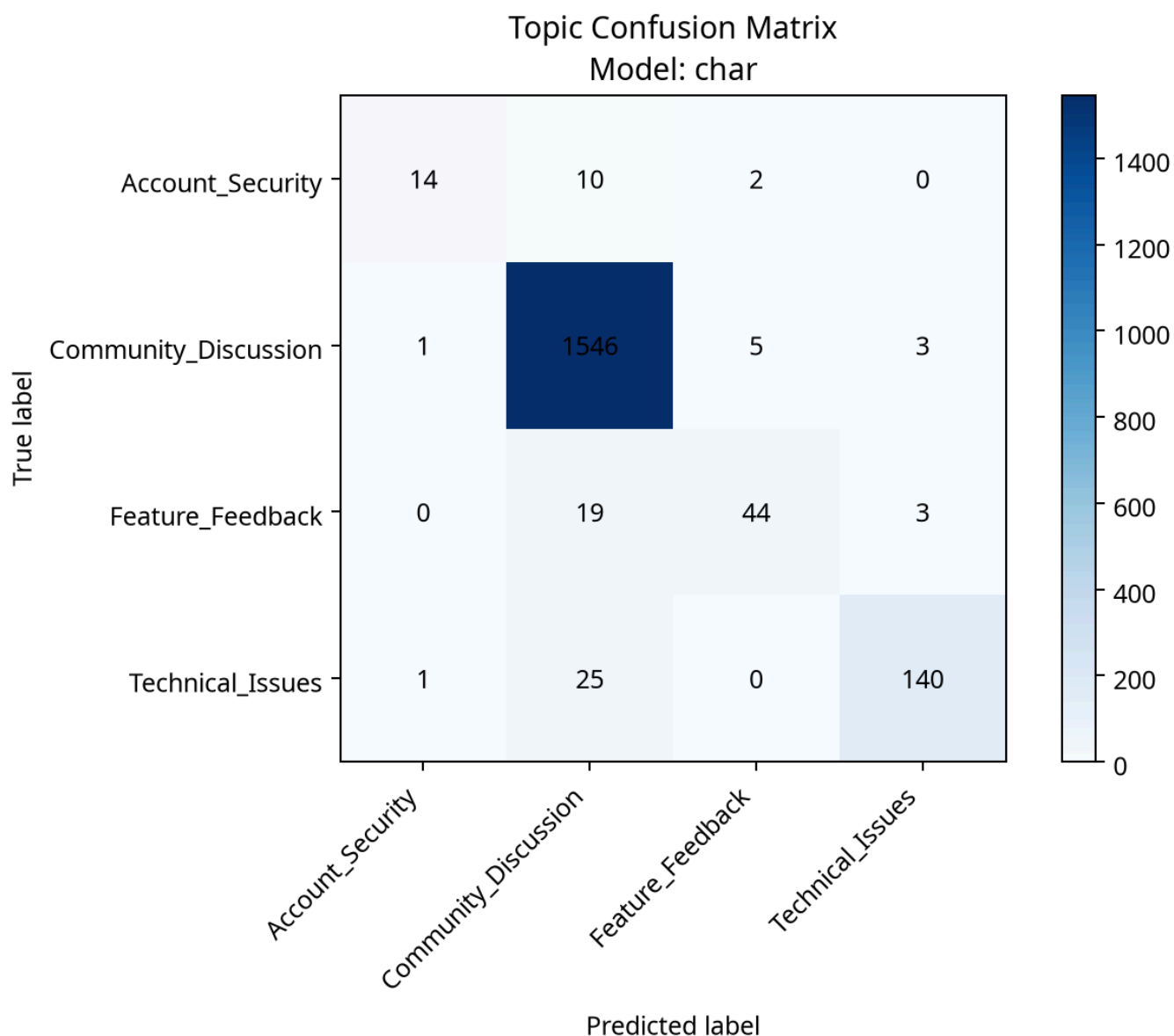
Final test metrics

Metric	Score
Accuracy	0.9619
Macro precision	0.9157
Macro recall	0.7607
Macro F1-score	0.8241
Weighted F1-score	0.9597

Per-class metrics

Class	Precision	Recall	F1-score	Support
Account_Security	0.8750	0.5385	0.6667	26
Community_Discussion	0.9663	0.9942	0.9800	1555
Feature_Feedback	0.8627	0.6667	0.7521	66
Technical_Issues	0.9589	0.8434	0.8974	166

Confusion matrix



3. Interpretation

The sentiment classifier achieved a macro F1-score of **0.6067**.

The topic classifier achieved accuracy of **0.9619** and macro F1-score of **0.8241**.

The topic accuracy is high partly because `Community_Discussion` is the dominant class. Macro F1 is therefore also reported.